



**Bulk ATAC-seq
fragments**



**scATAC-seq
fragments +
annotations**

1



atac-correct
Tn5 bias & SNR
95q-scaling

0

pseudobulk-fragments



B cell



T/NK



Mono

2



call-footprints
Identify & score FPs

3a

A C G T

match-motifs
JASPAR / HOCOMOCO

3b

De novo Discovery

motif-discovery
MEME/DREME
/STREME

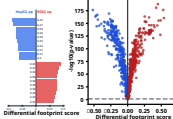


4

diff-footprints

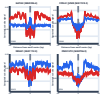


Replicate-aware
differential analysis



5

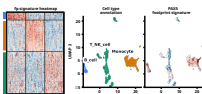
Bulk



plot-aggregate

6

Single Cell



find-signature-fp